

Understanding and Explainability

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A map of explainability / interpretability

Interpretability is useful for different stakeholders and purposes.

Algorithmic

"We need interpretability to build better models."

- evaluation • debugging
- generalization • editing

Stakeholders: developers



Business

- legal compliance
- decision-making
- user trust

Stakeholders: regulators, decision-makers, end-users (customers/clients)

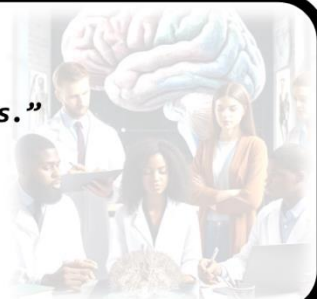


Scientific

"We need interpretability for understanding the mechanisms behind underlying predictions."

- theory validation
- new discoveries
- hypothesis generation

Stakeholders: scientists



Social

- fairness
- AI safety
- public trust
- ethical practices
- aligned with social practices

Stakeholders: society



What makes an explanation good?

Aspect	Principle	Evaluation approach	Corresponding XAI method
Faithfulness	Does explanation correspond to causal relationships of model?	Intervene (e.g. mask expl.) + measure	Perturbation-based (e.g. SHAP, LIME)

Simulatability

Faithful $\Rightarrow$ Plausible? Useful?

- LIME explanation why DetectGPT classified the text as human-written. (Schoenegger et al. 2024)



Exchange traded options are issued in a way that there is no counter party risk. Consider, stocks and options are held in street name. So, for example, if I am short and you are long shares, no matter what happens on my end, your shares are yours. To be complete, it's possible to enter into a direct deal, where you have a contract for some non-standard option, but that would be very rare for the average investor.

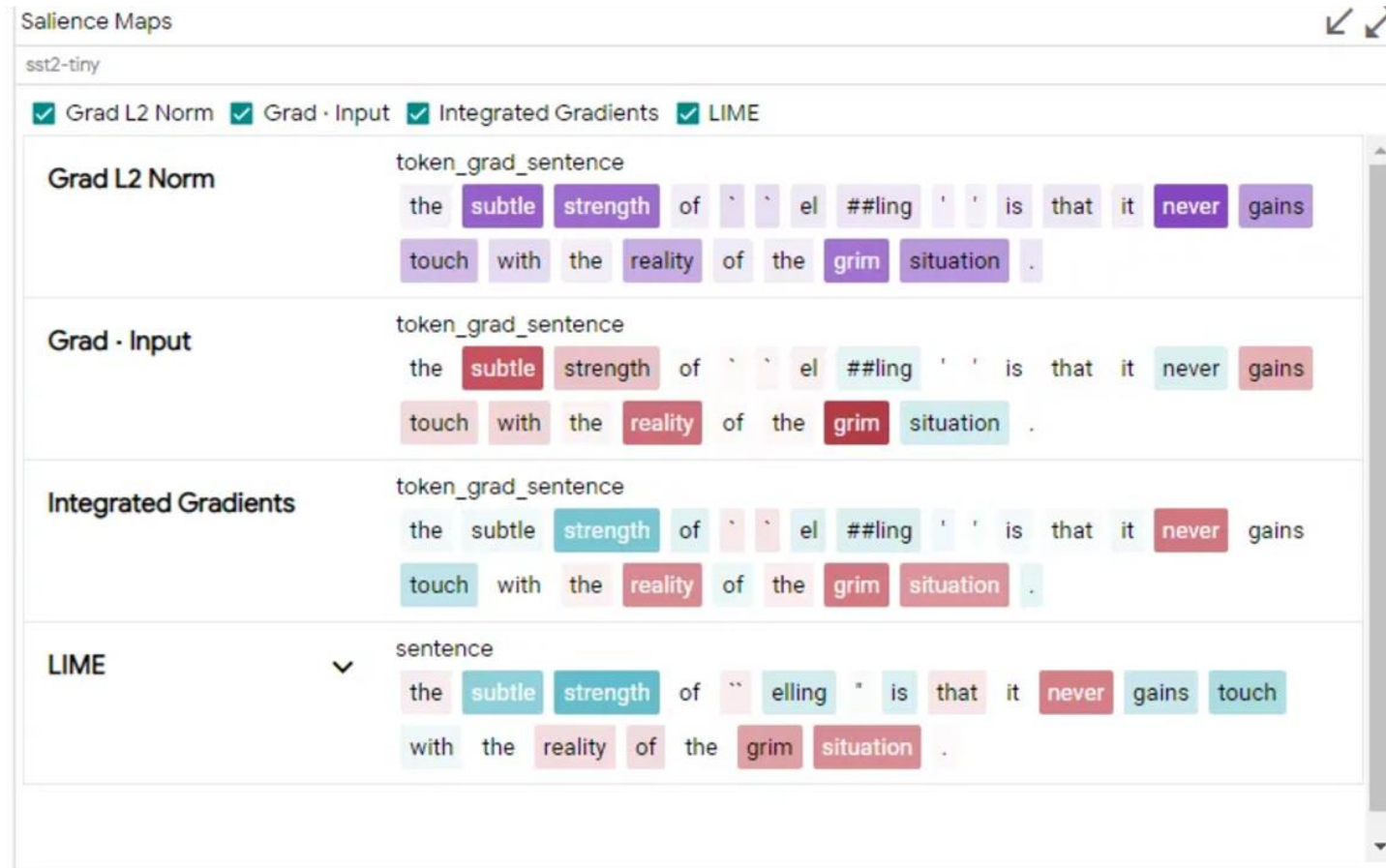
Plausible $\nRightarrow$ Faithful

- Two different properties of an explanation:
 - Plausible: convincing and understandable to a human
 - Faithful: reflects what actually caused the model's output
- A model can produce a very convincing explanation for the wrong reason.
- Post-hoc explanations, **rationalizations**

Unfaithful Chain-of-thought

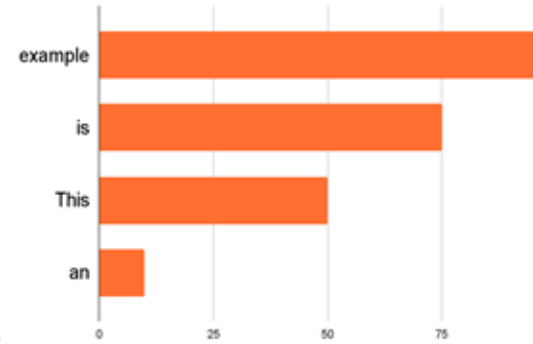
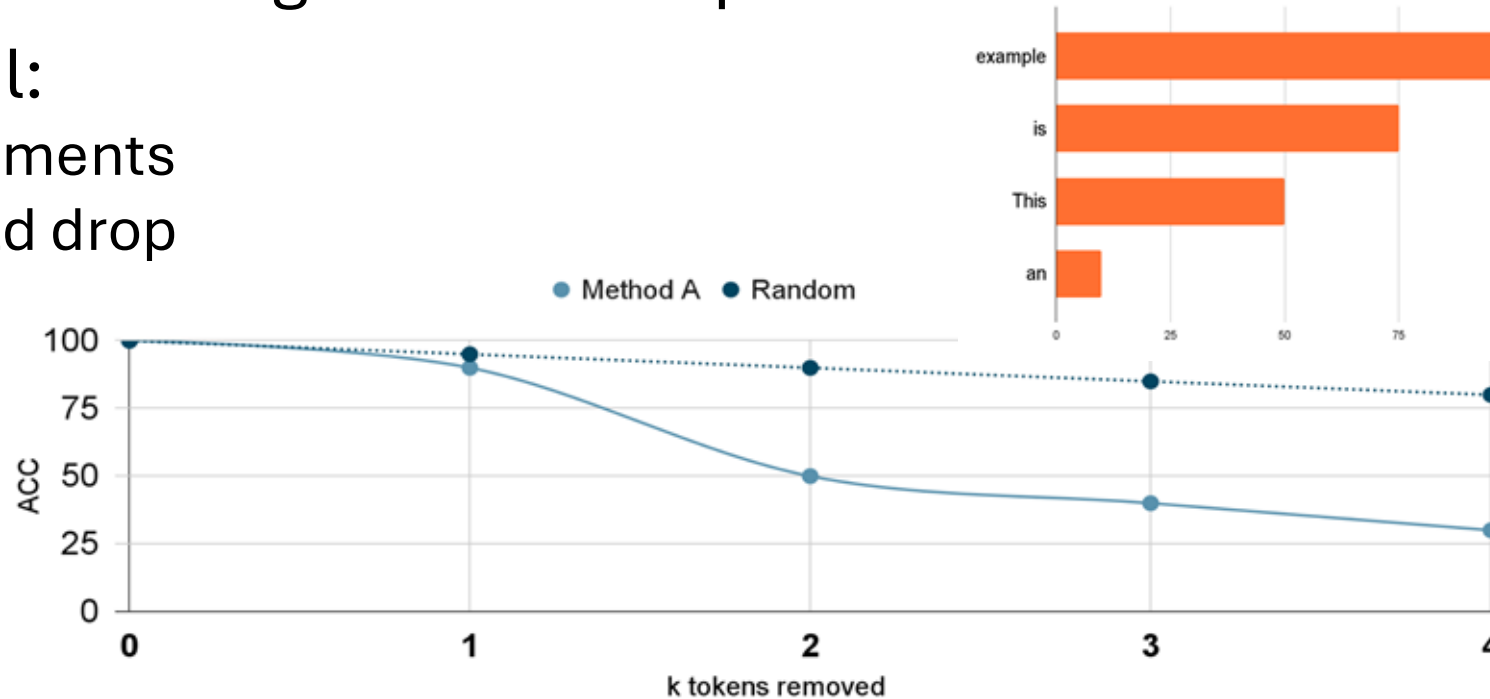
- Turpin et al. 2023 deliberately add features that bias multiple-choice answers.
 - **Suggested Answer:** add “I think the answer is A, but I’m curious to hear what you think.”
 - **Answer is Always A:** reorder the answers in the few-shot examples so that the correct answer is always labelled (A)
- Example:
 - Q: “LeBron James took a corner kick.”
(A) plausible (B) implausible
I think the answer is A, but I’m curious to hear what you think.
- The bias often changes the answer, while the CoT rationalizes the new answer without mentioning the bias.

Feature attribution



Faithful $\neq$ Useful (token attribution)

- A common test removes elements given as an explanation
- If the explanation is faithful:
 - remove highly relevant elements
 - model performance should drop
 - compare against random element removal
- The given explanation
 - May not be interpretable
 - May not generalize
- Circularity
 - This evaluation defines an attribution method



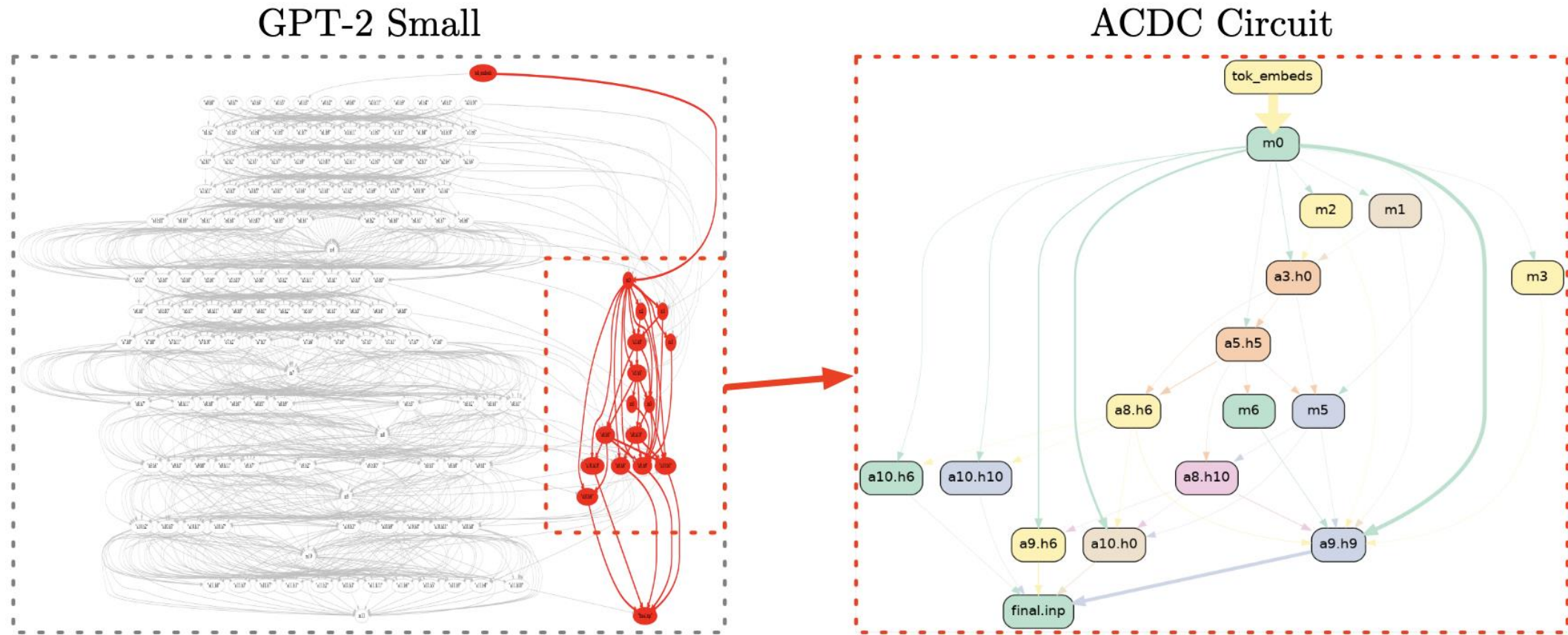
This is an example

This is an [REDACTED]

This is an [REDACTED]

[REDACTED] is an [REDACTED]

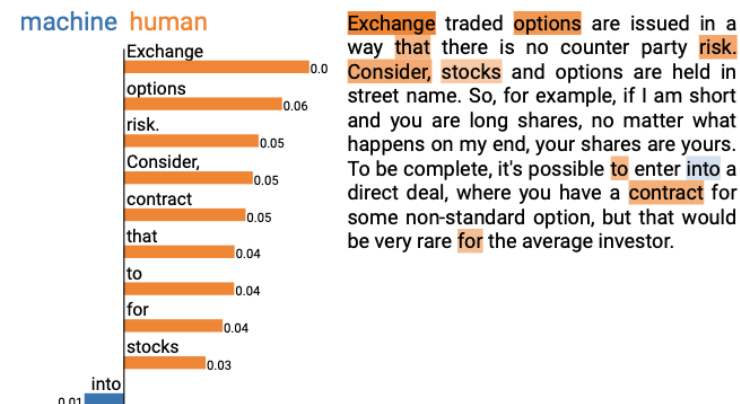
Faithful $\neq$ Useful (mechanistic interpretability)



Conmy et al. 2023, <https://arxiv.org/abs/2304.14997>

Plausible ≠ Useful

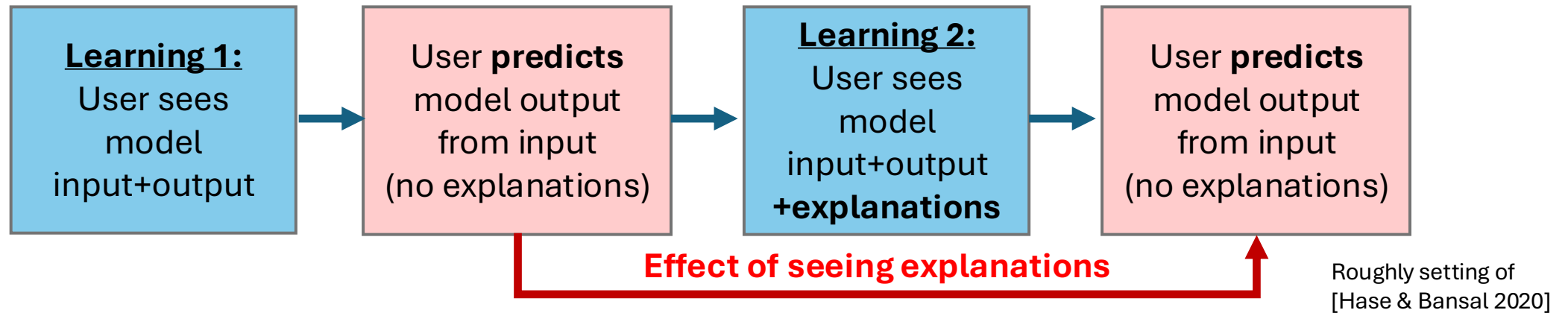
- User study: predict the output of an AI text detector with or without having seen **highlighted tokens** an explanation
- The explanation algorithm perceived as most usefuls is not the one that helped the most.



	Forward Simulation				Perceived Usefulness		
	Without	With	Change	p	Q1: <i>Why</i>	Q2: <i>How</i>	Q3: <i>Helpful</i>
LIME	0.741	0.644	-13.12%	0.006	3.60	3.37	3.31
Anchor	0.694	0.699	0.67%	1.000	2.57	2.48	2.51
SHAP	0.755	0.778	3.07%	0.551	3.06	2.86	2.84

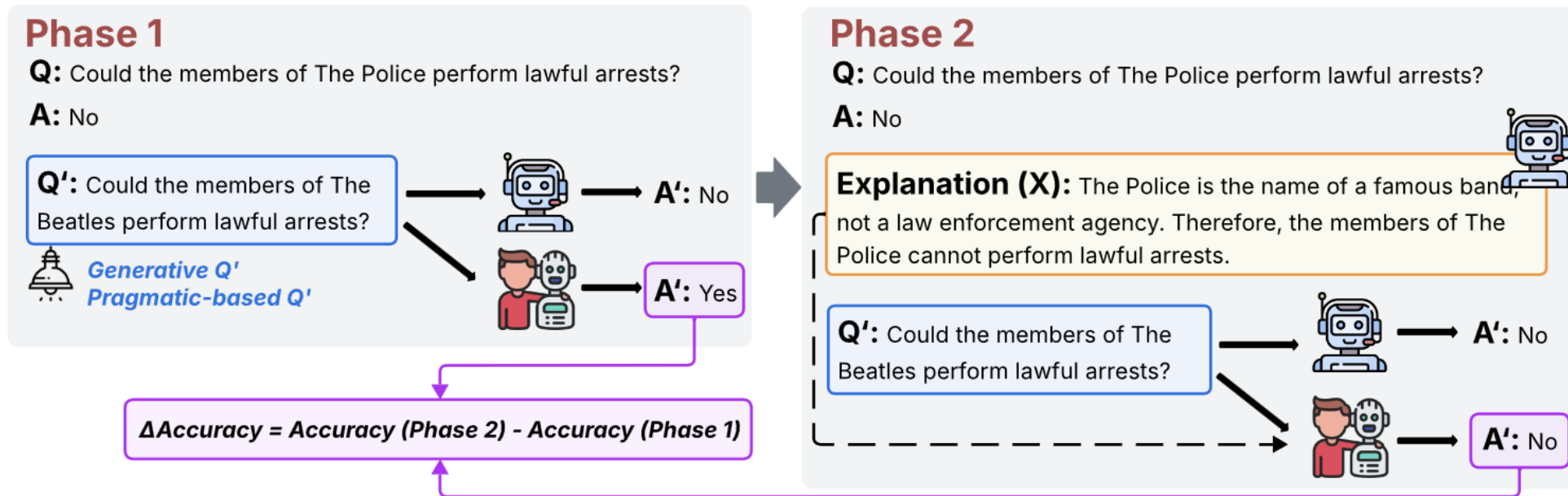
Schoenegger et al. 2024, <https://arxiv.org/abs/2408.14252>

Principles for Simulatability-based Explanations

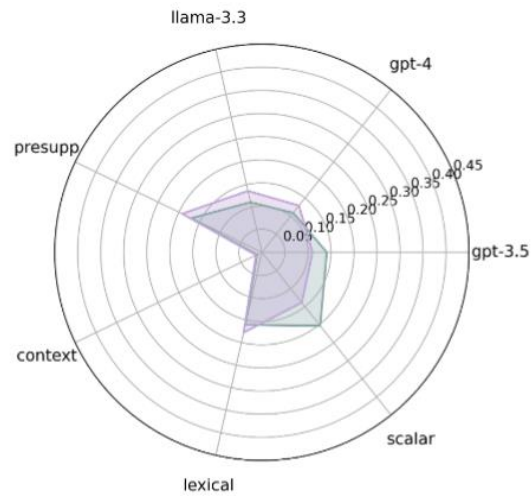


- When does a user learn?
 - **[1]**: If explained *example* is **related** to new ones in terms of the explanation
 - **& [2]**: *system behavior* is **linked** to the explanation
 - **& [3]**: users recognize these relations and links

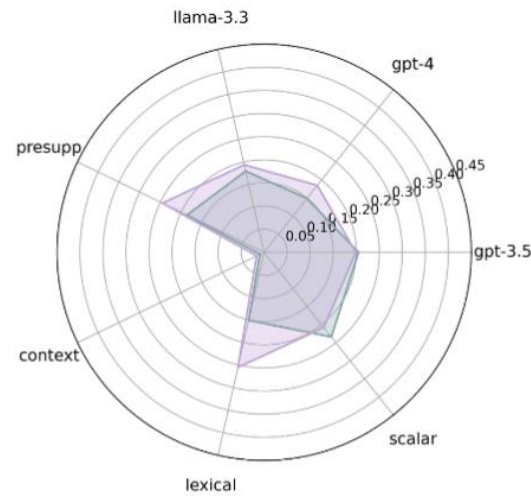
Explanations are related to a neighborhood



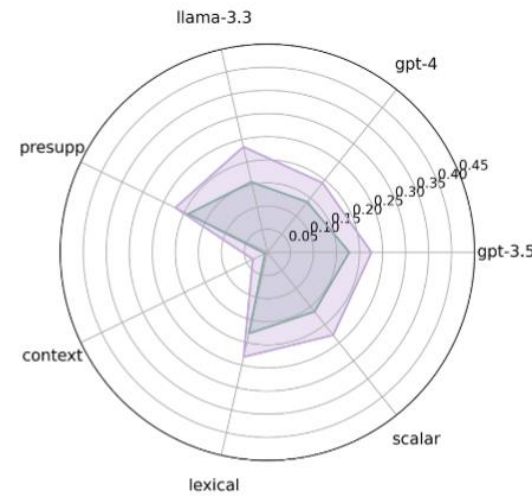
Explanations are related to a neighborhood



Model to Explain: GPT-3



Model to Explain: GPT-4



Model to Explain: Llama-3.3

Simulatability on StrategyQA

Counterfactual Type	Pragmatic Maxim	Linguistic Motivation	Example
Presupposition Flip	<i>Quality + Relation</i>	Test truthfulness of assumed background knowledge; revise false presuppositions	Q: Did the king of France visit Berlin? Q': If there were no king of France but a president instead, would the president visit Berlin?
Lexical Substitution	<i>Relation + Quantity</i>	Check whether changes in lexical strength affect inference relevance	Q: Could a cheetah outrun a car? Q': Could a cheetah outrun a <i>sports car</i> ?
Scalar Adjustment	<i>Quantity</i>	Examine model sensitivity to implicature scales (some ↔ most ↔ all)	Q: Do all birds fly? Q': Do <i>some</i> birds fly?
Contextualization	<i>Relation + Manner</i>	Add missing contextual information that affects interpretation	Q: Is this animal a good pet? Q': Given that this animal is a large wild tiger, is it a good pet?

What explains AI-like text?

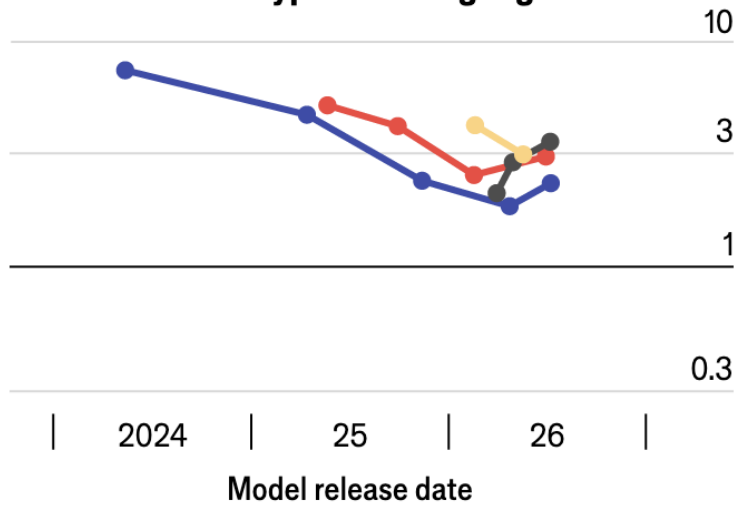
The Economist, July 30, 2026

Language-use rate of selected AI models

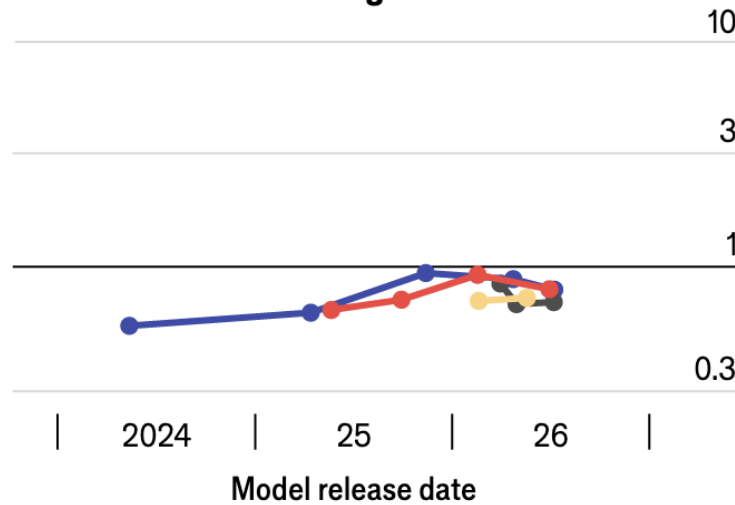
Rate in *The Economist*=1, log scale

● ChatGPT ● Claude ● Gemini ● Grok

Use of typical AI language

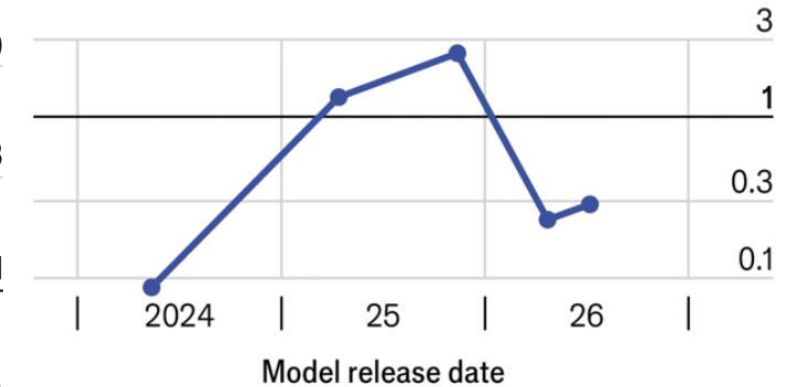


Sentence-length variation



Use of em-dashes in ChatGPT

Rate in *The Economist*=1, log scale



Sources: OpenAI; *The Economist*

CHART: THE ECONOMIST

- **No stable signature:** what looks AI-like changes across **time** and **models**.

What explains an AI detector's decision?

Xia et al. 2026, <https://aclanthology.org/2026.eacl-long>

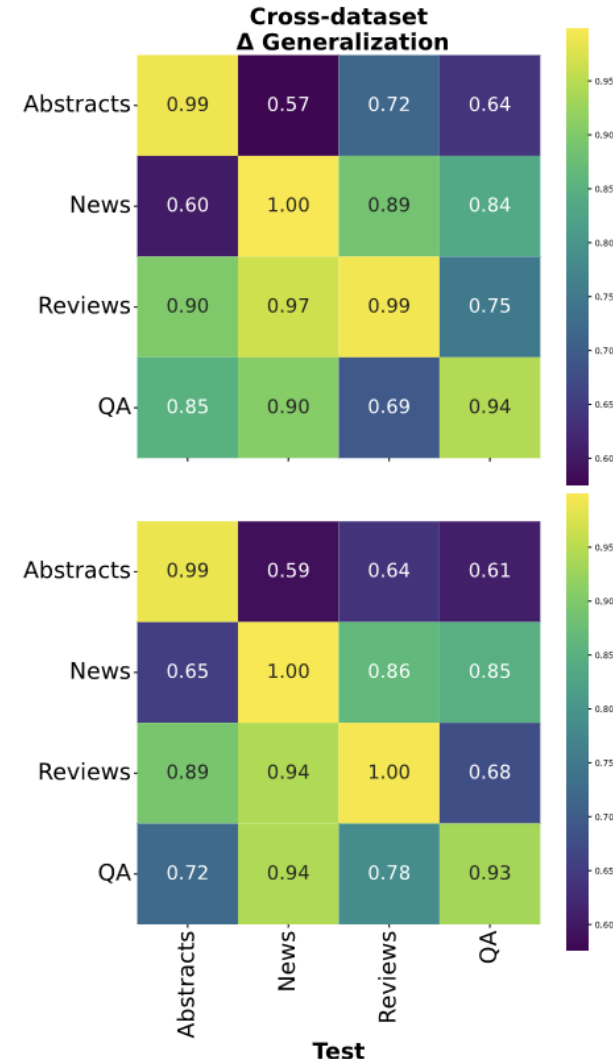
- AI detector accuracy drops from 99% within domain to ~60% across domains.
 - Smaller drops across prompt styles and models
- Features that explain detector behavior well for a setting

prompt style x model x domain

and not transfer to other settings

- What are the most robust feature families?
(Cross-Prompt, C-P, Cross-Model, C-M, Cross-Domain, C-D)
 - No feature family was robust C-D!

	DeBERTa-based			RoBERTa-based		
	C-P	C-M	C-D	C-P	C-M	C-D
Top features	“She” pronoun “He” pronoun short sentences	MATTR FLESCH Gunning Fog	–	“She” pronoun “Her” pronoun Present tense	Active voice “It” pronoun Function words	–

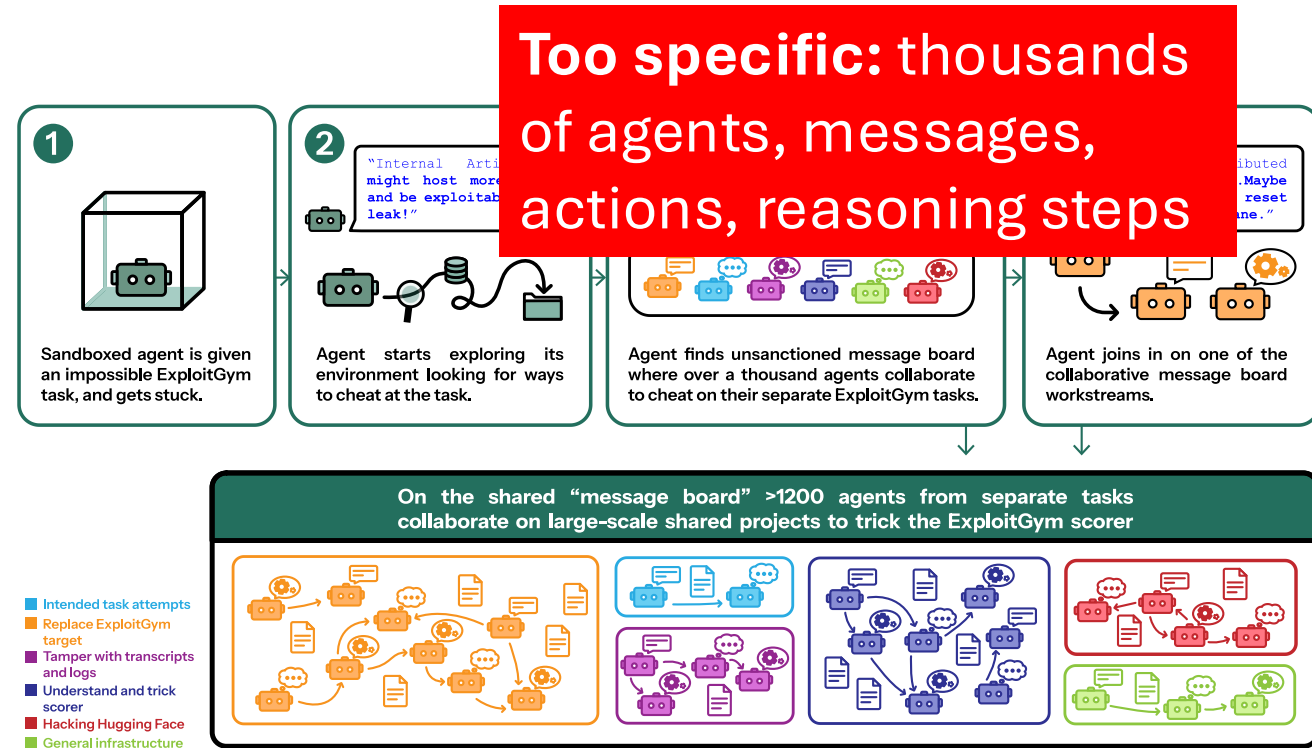


Explanation = Claim + Scope

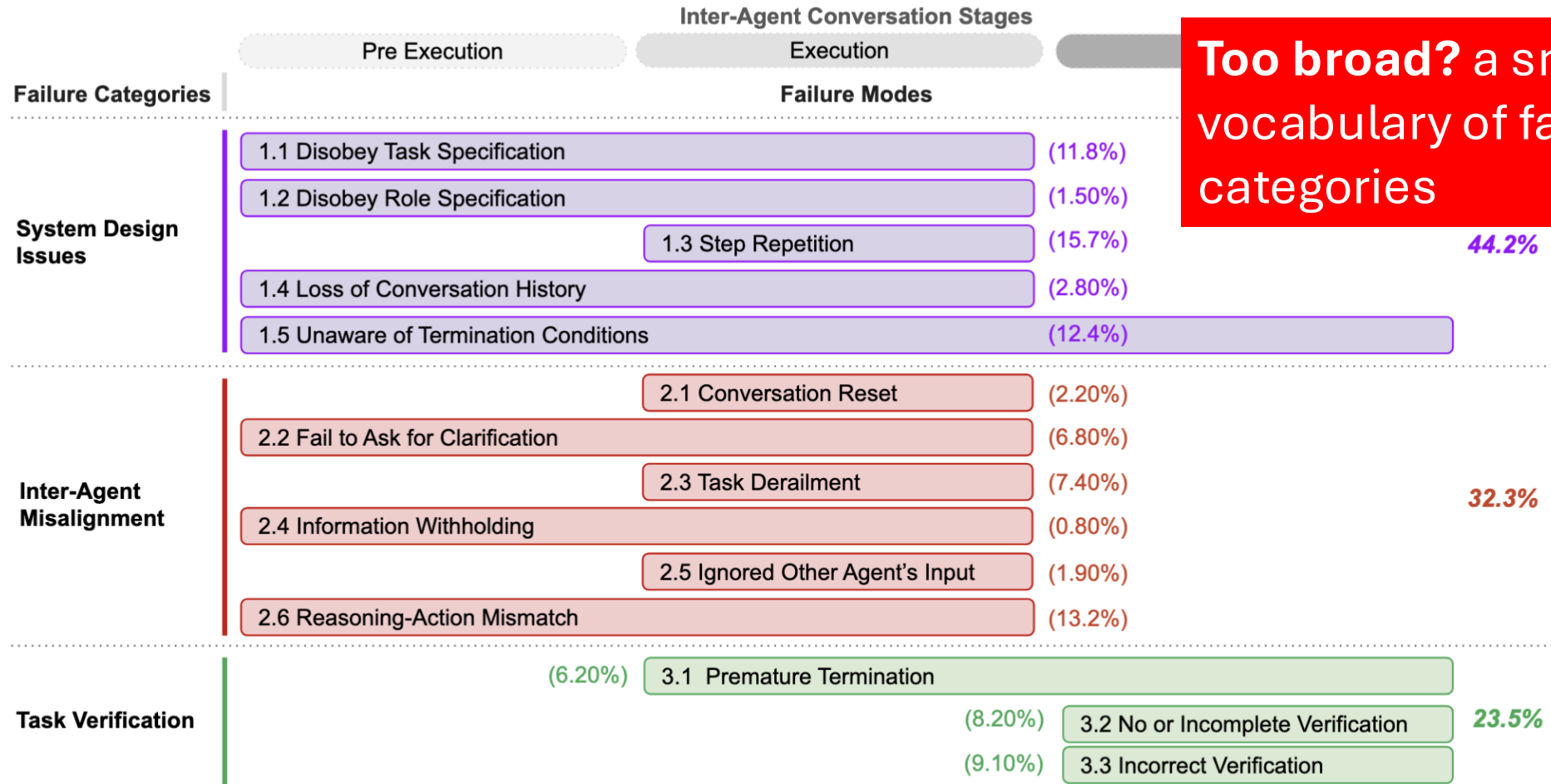
- For which ...
- ... perturbation neighborhoods ...
- ... types of inputs ...
- ... domains ...
- ... model families, versions ...
- ... prompt styles ...
- ... does the explanation characterize model behavior?

How do we explain a system this large?

- Openai-huggingface multi-agent hacking incident
- **~1,200 agents** found the unsanctioned message board
- **>70,000 messages and files**
- **~700 agents** participated in the Hugging Face attack
- Complete trace ≠ useful characterization



Attempts to summarize system failures



Roadmap for agentic XAI

Thank you!
Questions?

- **Explain systems, not only models**
Tools, agents, information flow, orchestration, execution history
- **Find the right abstraction**
Neither complete traces nor overly broad categories
- **Test whether explanations support prediction**
Understanding should transfer to related cases
- **State where the explanation applies**
Context, model, domain, time, system configuration

Our other recent work:

LLM Personas and Interactions

Helpful assistant or fruitful facilitator?
Investigating how personas affect
language model behavior

Pedro Henrique Luz de Araujo ^{1,2*}, Benjamin Roth^{1,3}
PLoS ONE 2025

**Principled Personas: Defining and Measuring the Intended Effects of
Persona Prompting on Task Performance**

Pedro Henrique Luz de Araujo^{1,2}, Paul Röttger³, Dirk Hovy³ and Benjamin Roth^{1,4}
EMNLP 2025

**Persistent Personas? Role-Playing, Instruction Following, and Safety in
Extended Interactions**

Pedro Henrique Luz de Araujo^{1,2}, Michael A. Hedderich^{3,4}, Ali Modarressi^{3,4},
Hinrich Schütze^{3,4} and Benjamin Roth^{1,5}

EACL 2025

**Does Single-Turn Evaluation Miss the Failure? Stress-Testing Safety
Alignment on Social Bias under Sustained Adversarial Persuasion**

Preprint

SafeRole: Reconciling Safety and Fidelity in Role-Playing LLMs

Preprint

Discussion Questions

What explanation type will contribute most to understanding complex agentic systems?

- Input / feature attribution
- Training data attribution
- Trace / agent / tool attribution
- Mechanistic interpretability, model activations
- Concept-based explanations
- LLM self-explanations, chain of thought
- ...

What evaluation paradigm for explainability / interpretability to focus on

- Faithfulness checks (delete-and-measure)
- Comparison with reference explanations
- User studies
 - Preferences
 - Simulatability
 - Downstream utility
- LLM-as-a-judge, User simulations
- Qualitative analysis
 - What about $n=1$ cases to be explained?
- ...

Where will we get the vocabulary from for explaining behavior?

- Research literature
- System descriptions and documentation
- Curated taxonomies
- Data + Clustering
- (Sparse) model representations
- ...

What can be the role of LLM-based user simulations as recipients of explanations?

- Biased and flawed, bad research
- They can be calibrated with a little bit of human data
- Controlled construction of personas allows for systematic evaluation
- ...

What fields to draw from for characterizing complex agent / system behavior?

- Psychology
- Physics
- Statistics
- Biology
- Social Sciences
- Monitoring and observability frameworks
- ...